

Outline:

1. Why Version Control?

- Teamwork, backup, history.
- Problems without version control.

2. What is Git & GitHub?

- Git = tool (local).
- GitHub = platform (cloud).

3. Basic Git Workflow

- git init, git add, git commit.
- git push and git pull.

4. Branching Basics

- master/main vs develop.
- git checkout -b feature-xyz.

5. Collaboration on GitHub

- Forks, Pull Requests, Issues.

6. Demo/Practice

- Create repo → clone → commit → push.

PowerPoint 2 – Consuming APIs in C#

Outline:

1. What is an API?

- Analogy: Restaurant menu (request → response).
- REST APIs & HTTP basics.

2. HTTP Verbs

- GET, POST, PUT, DELETE (CRUD mapping).

3. Response Codes

- 200 OK, 201 Created, 400 Bad Request, 404 Not Found, 500 Internal Server Error.

4. Consuming APIs in C#

- Using HttpClient.
- Demo calling a public API (e.g., OpenWeather, GitHub API).

5. Error Handling

- Handling unsuccessful responses.
- Try/catch + response.EnsureSuccessStatusCode().

6. Demo/Practice

- Create console app → call API → print results.

PowerPoint 3 – Creating Your Own Web API in C#

Outline:

1. Why Build APIs?

- Share data across apps.
- Example: Mobile app + website using same backend.

2. ASP.NET Core Web API Basics

- Project setup (dotnet new webapi).
- Structure: Controllers, Program.cs, appsettings.json.

3. Creating Endpoints

- GET, POST, PUT, DELETE.
- In-memory data source.

4. Request & Response

- Route parameters.
- Body (JSON input).
- Returning proper status codes.

5. **Testing APIs**

- Swagger (built-in).
- Postman.

6. **Demo/Practice**

- Create simple ProductsController.
- Test with Swagger.